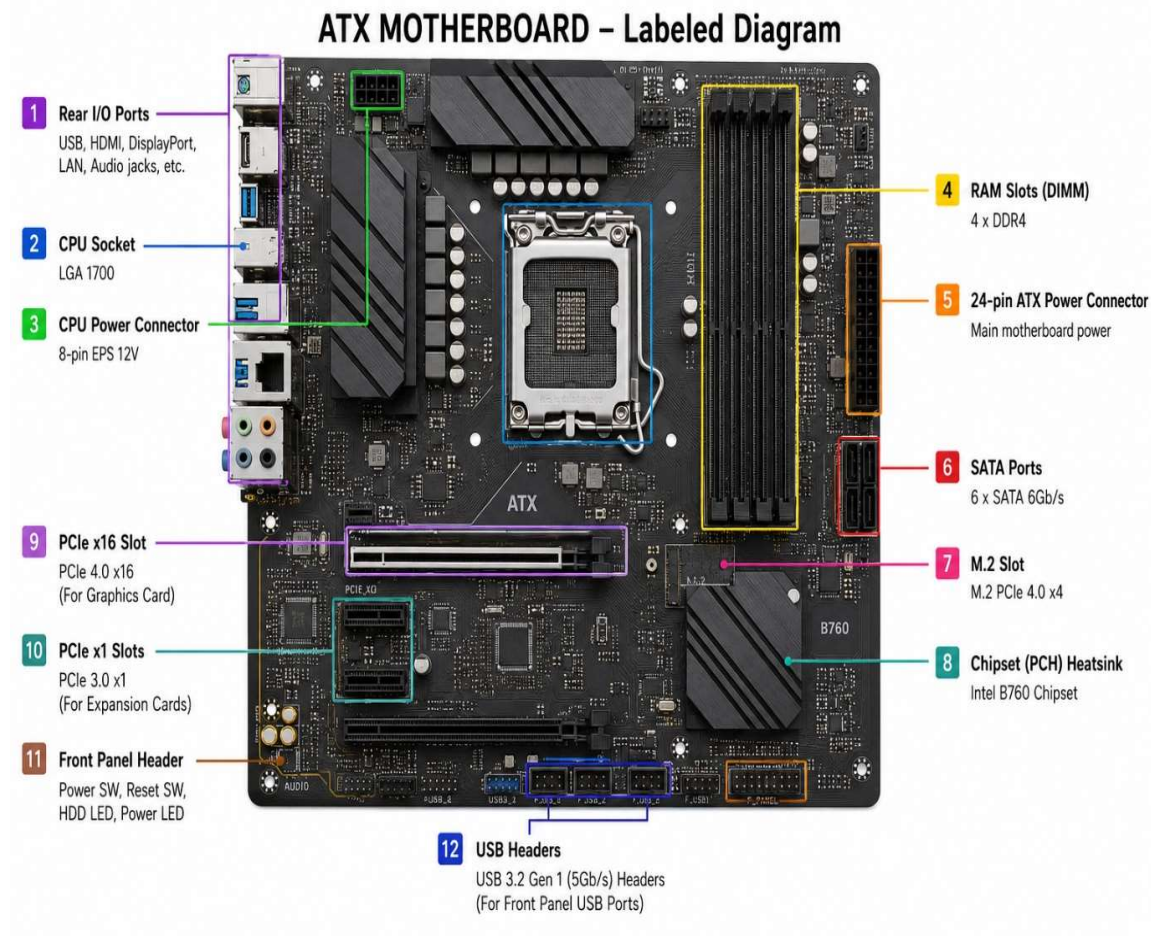


1. Draw a labelled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type	Compatible CPU Model	Manufacturer
LGA2066	Intel Core i9-10980XE	Intel
FM2+	AMD A10-7890K	AMD
LGA775	Intel Core 2 Quad Q9650	Intel

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

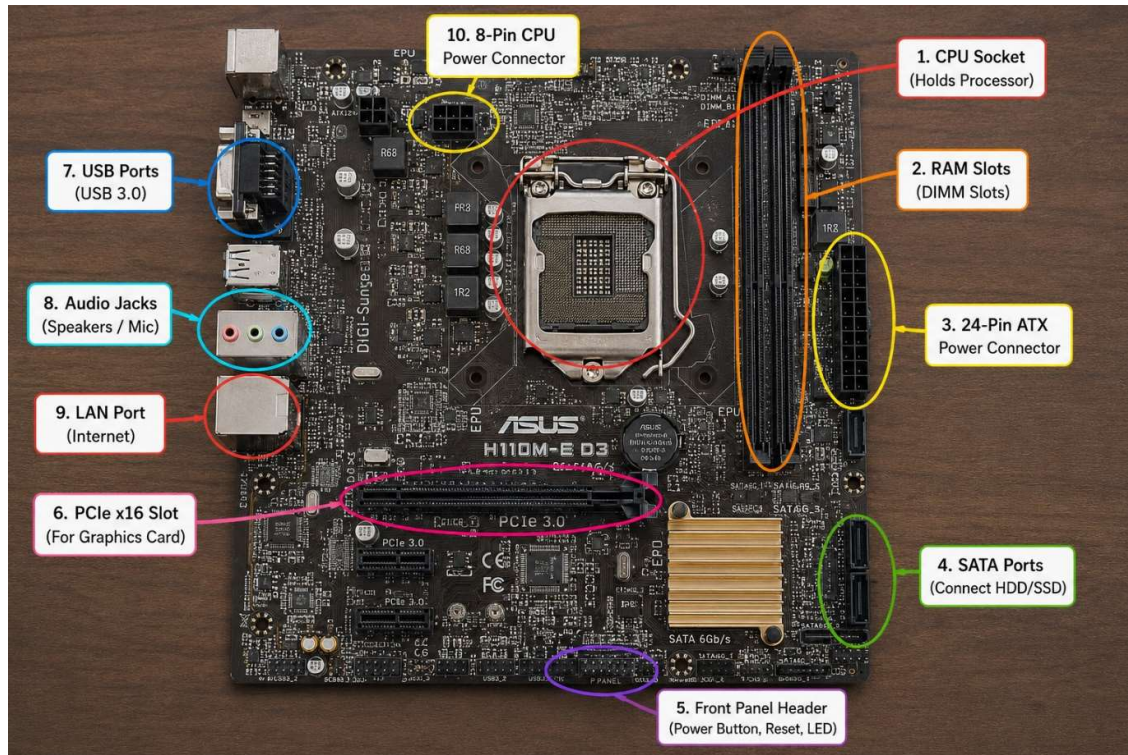
❖ Functions of the CPU and Chipset:

- performs calculations, processes instructions, and controls most of the computer's operations. It is often called the "brain" of the computer because it handles the actual work of running programs and processing data.
- The chipset is a group of electronic components on the motherboard that manages communication between the CPU and other hardware devices, such as RAM, storage drives, USB ports, and graphics cards. It helps ensure that data moves smoothly between different parts of the computer.

❖ How They Work Together When Opening Spotify

- When you open Spotify, the CPU receives the command and starts running the application's instructions. It loads the program into memory, processes user inputs, and manages tasks such as displaying the interface and playing music.
- At the same time, the chipset helps transfer data between the CPU, RAM, and storage device. For example, it assists in retrieving Spotify's files from the SSD or HDD and sending them to the CPU and memory. When music is played, the chipset also helps communicate with the audio hardware so that sound can be heard through speakers or headphones.
- In simple terms, the CPU does the thinking and processing, while the chipset acts like a traffic manager, helping all the computer components communicate efficiently. Together, they allow applications like Spotify to open quickly and play music smoothly.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.



1. CPU Socket

The CPU socket is where the processor is installed on the motherboard. It connects the CPU to the computer so it can process data and run programs.

2. RAM Slots

RAM slots hold the computer's memory modules. They help the computer run multiple programs smoothly and quickly.

3. 24-Pin ATX Power Connector

This connector provides main power to the motherboard from the power supply. Without it, the computer cannot turn on.

4. SATA Ports

SATA ports connect storage devices like HDDs and SSDs to the motherboard. They allow the computer to store and access data.

5. Front Panel Header

This header connects the power button, reset button, and LED lights from the computer case. It lets you control the PC from the front panel.

6. PCIe x16 Slot

The PCIe x16 slot is mainly used for installing a graphics card. It provides high-speed communication between the card and the motherboard.

7. USB Ports

USB ports are used to connect devices such as keyboards, mice, printers, and flash drives. They allow data transfer and power delivery.

8. Audio Jacks

Audio jacks connect speakers, headphones, and microphones to the computer. They handle sound input and output.

9. LAN Port

The LAN port connects the computer to the internet using an Ethernet cable. It provides a stable and fast network connection.

10. 8-Pin CPU Power Connector

This connector supplies power directly to the CPU. It helps the processor work properly, especially during heavy tasks.